Introduction
to
Computer & its Architecture

What is Computer?

- A computer is a programmable electronic device that accepts raw data as input and processes it with a set of instructions (a program) to produce the result as output.
- The term "computer" is derived from the Latin word "computare" which means to calculate.
- A computer is designed to execute applications and provides a variety of solutions through integrated hardware and software components.

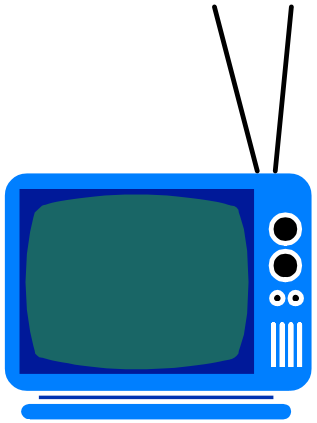
Which one is the computer?



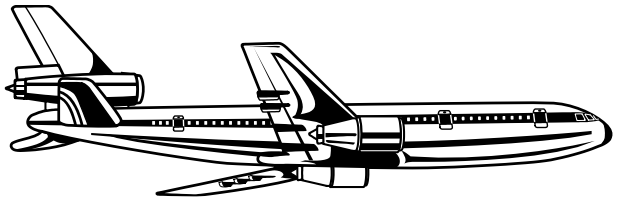
Rock



Calculator



Television



Modern Airplane



Washing Machine



Computer Workstation

Is a rock a computer?



- Does not act or process
- Takes no input and produces no output

- Computers must be able to handle *input* and *output*

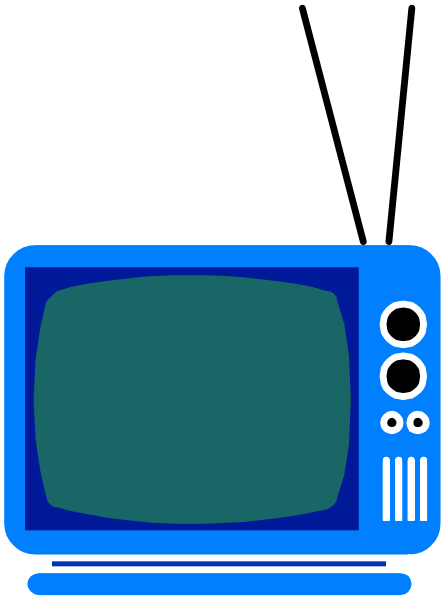
Is a washing machine a computer?



- Input: dirty clothes
- Output: clean clothes
- Does not handle information
- The microprocessor unit in your washing machine controls sensors, valves and actuators responsible for controlling the parts that clean your clothes

- Computers input and output as *information*

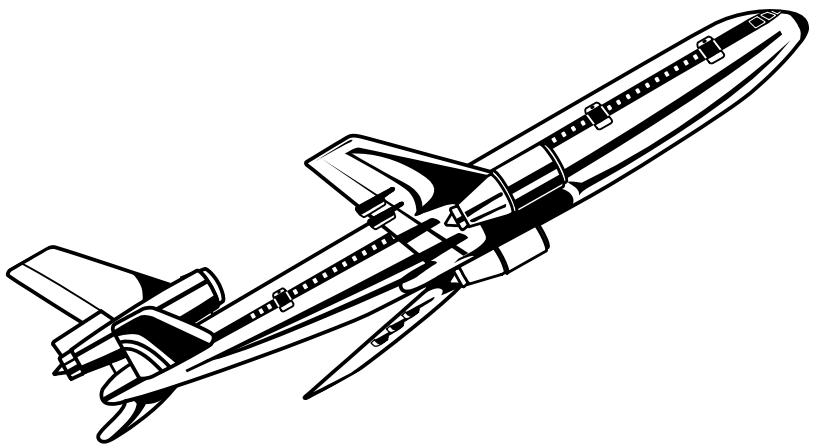
Is a television set a computer?



- Input: information from cables or radio waves
- Output: information as sound and picture
- Does not process information

- Computers *process* information by computing new results and answering queries

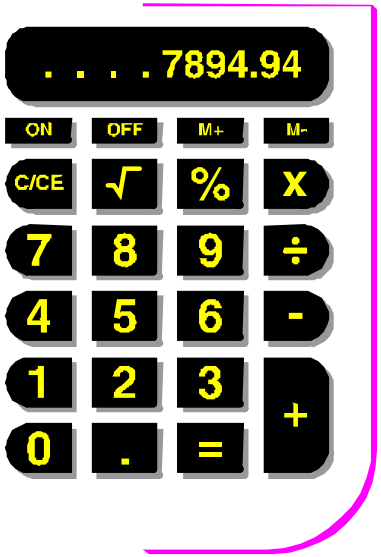
Is a modern airplane a computer?



- Input: information from radio waves
- Output: manipulations to the airplane
- Can only handle specific information necessary for flight control

- Computers are *general purpose* because they can perform many different tasks

Is an ordinary calculator a computer?



- Input: numbers and mathematical operations
- Output: answer
- Handles any numeric task
- Cannot remember which buttons are pressed

- Computers are *programmable* so they can remember sequences of operations

Definition of a Computer

- a general purpose,
- programmable,
- information processor
- with input and output



Characteristics of Computer

1. Speed
2. Accuracy
3. Reliability
4. Versatility
5. Storage Capability
6. No I.Q
7. No Feelings

Speed...

- Process data at an extremely fast rate
- Computer perform task in minutes that may take days if performed manually
- A modern computer can execute millions of instructions in seconds.



Accuracy...

- Work with the highest Accuracy
- Computers can perform all calculations and comparison accurately , provided the Hardware does not malfunction.
- Human Errors are like wrong command, inaccurate data input etc



Storage Capability…

- Unlike human being Computer can store huge amount of data
- Loss or Deletion of data is done by User
- The computer memory is small and it can hold only a certain amount of information, therefore the data is stored on storage devices such as magnetic tape or disk.
- Todays flash memory sticks (Such as pen drives) of 3-5 cms long can store upto 128 Gigabytes (256 GB) data.



Reliability…

- At hardware level, it does not require any human intervention between its processing operations.
- Computers are immune to tiredness, boredom and fatigue . Therefore , more reliable than human beings.

...Characteristics of Computer...

- Free from "Tiredness"
- Free from Lack of

  "Concentration"





Versatility...

- Can perform all kind of job simultaneously
- They can work in areas where human brain can make error,For instance, observing a motion of very fast moving articles.



No IQ...

- Not a thinking machine
- Not Intelligent
- Never does a task as it own
- Follow Human Instructions
- Only perform what it is programmed to do :

  nothing more & nothing less

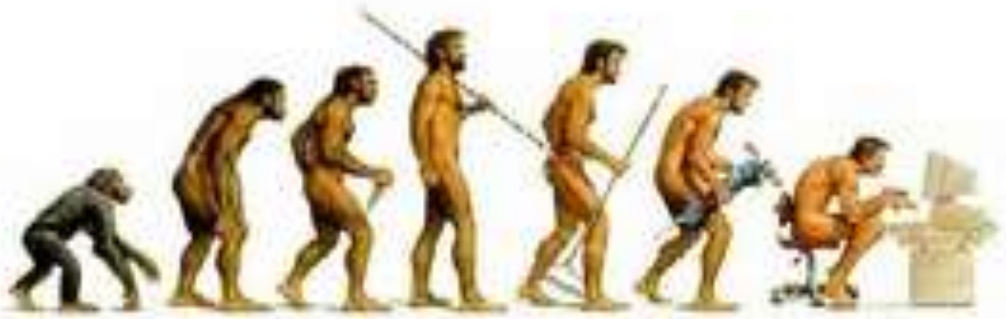


...Characteristics of Computer...

No Feeling…

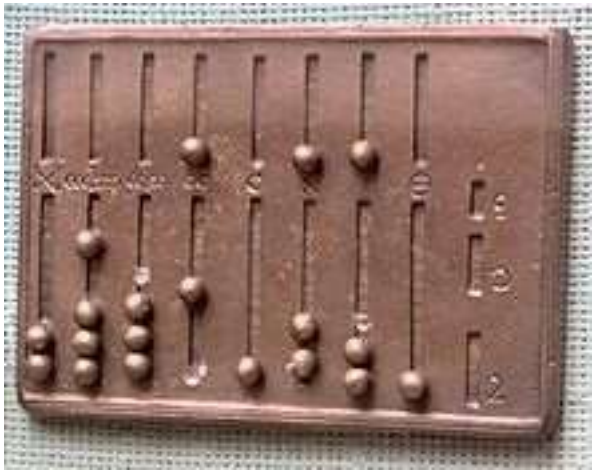
- No taste
- No emotions
- No knowledge

Evolution of the Computer



ABACUS

- Abundant Beads, Addition and calculation Utility System
- Mechanical Device
- Basic Arithmetic Operation



Roman Abacus



Chinese Abacus



Russian Abacus

Russian Abacus

Napier bones(1614)

- Specially constructed Board with moving roads
- By John Napier
- Useful for Large Calculation like Product and quotients



NAPIER'S BONES

Russian Abacus

Pascaline(1642)

- Multiplication and division through repeated addition and subtraction
- Useful for accounting purpose



Pascaline

Analytical engine(1834)

- By Charles Babbage(Father of computer)
- Mechanical computer
- Concept of input,output,storage
- Used for commercial purpose



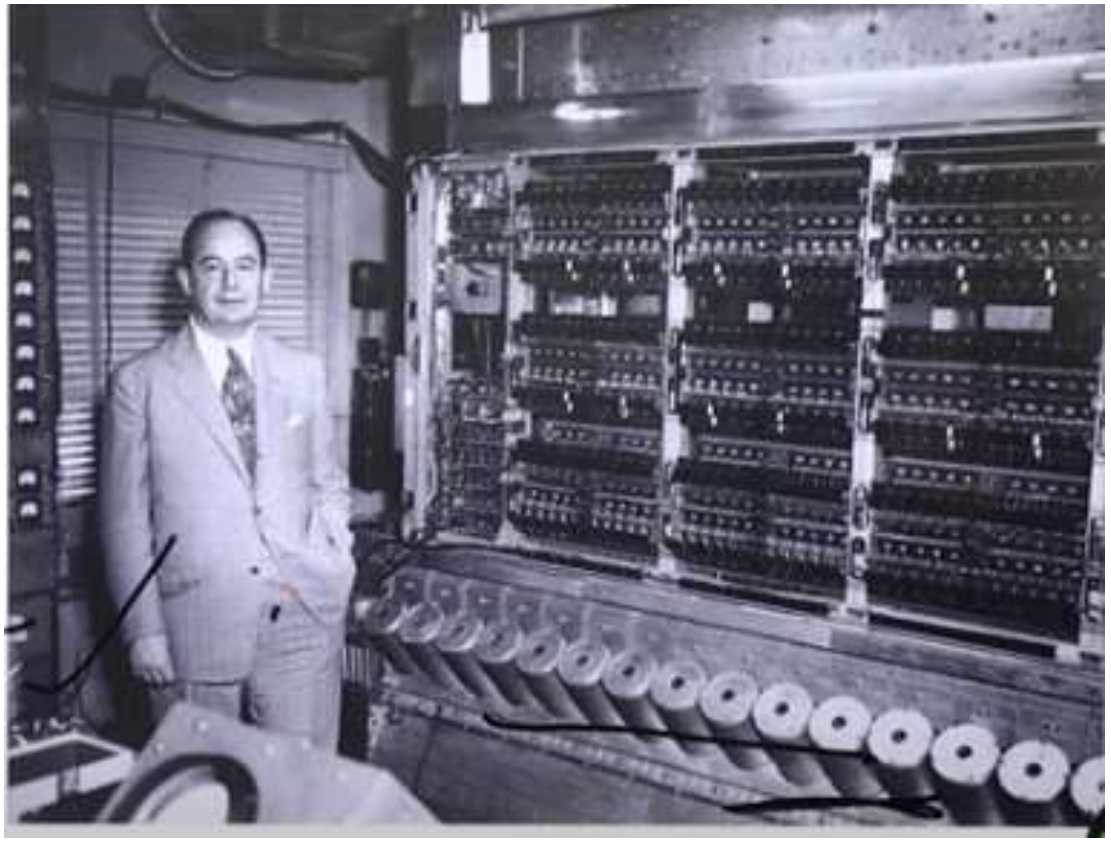
Census Tabulator(1890)

- By Harman Hollerith
- Tabulation Machine
- Uses Punched card



Turing Machine(1937)

- By Alan Turing
- Programmable device
- Uses Algorithm
- Uses Punched card



Generation of Computer











Generations of Computer

- **'Generation' : stage of technological development or innovation.**

- **According to the kind of 'processor' installed in a machine, there are five generations of computer**

  - **The First generation**
  - **The Second Generation**
  - **The Third Generation**
  - **The Fourth Generation**
  - **The Fifth Generation**

The First Generation

- 1940-1956
- Vacuum Tube
  - Heat
  - Burnout
- Machine language (0/1)
- Input
  - Punched cards
- Output
  - Printout



Characteristics of 1st Generation Computers

- Computers big and clumsy
- Electricity consumption is high
- Hardware failure occurred regularly - computers not very reliable
- Large air conditioners was necessary because the computers generated heat
- non-portable, lacked in versatility & speed
  - *Batch processing (one problem at a time)*

The First Generation

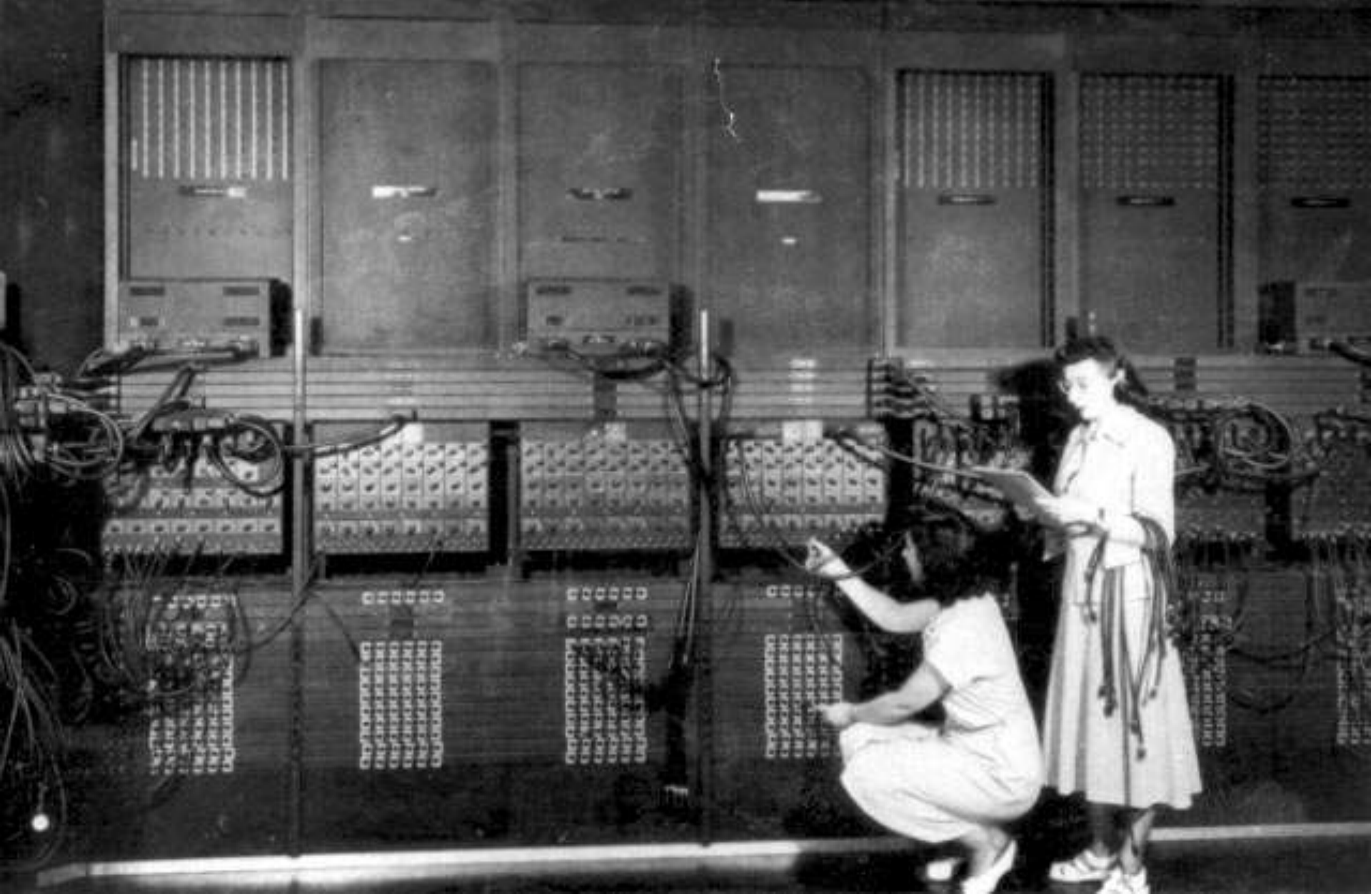
- Some of more Example of 1st generation computers are

ENIAC (Electronic Numerical Integrator And Computer)

EDVAC (Electronics Discrete Variable Automatic Computer)

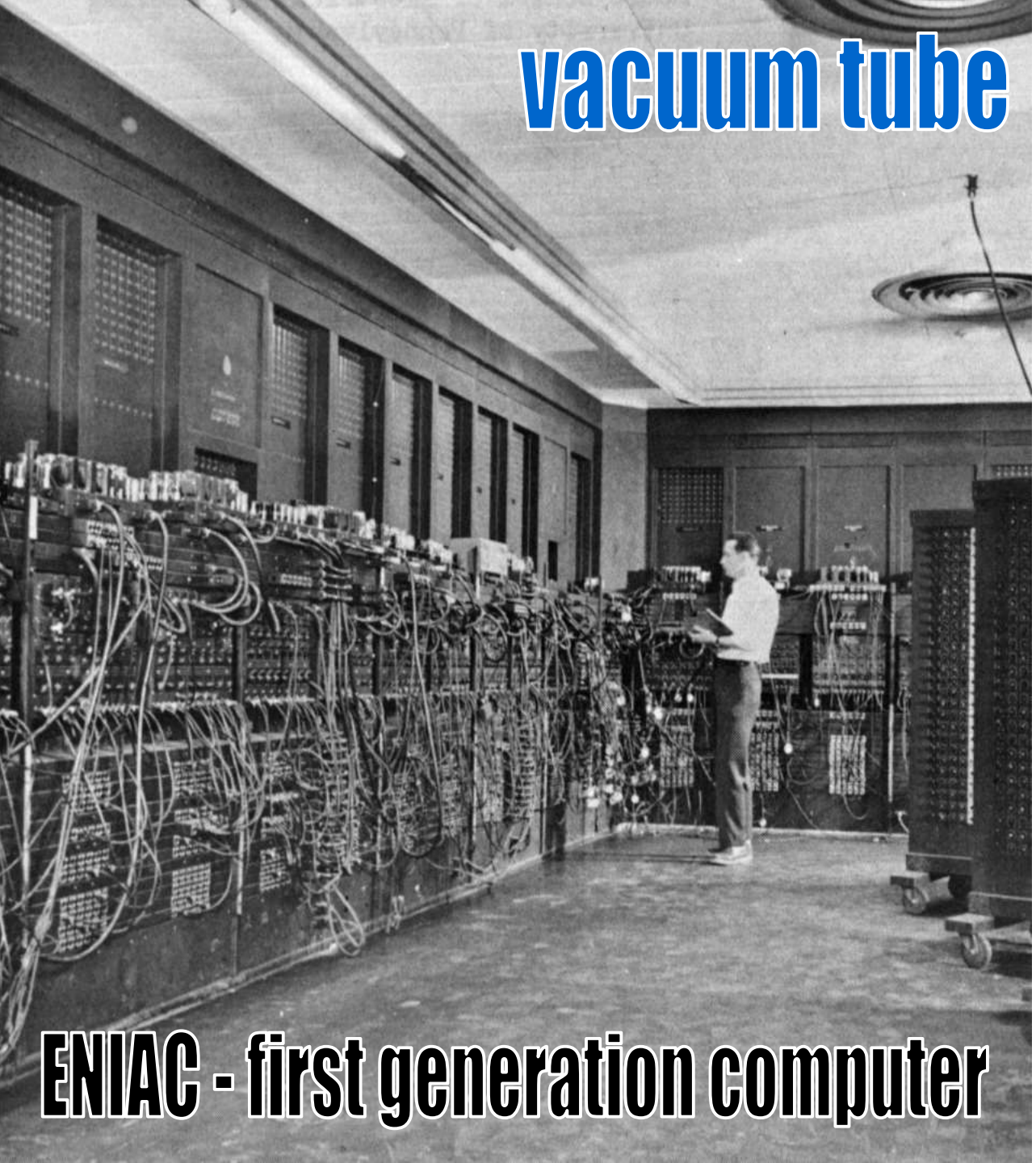
EDSAC (Electronics Delay Storage Automatic Calculator)

UNIVAC-1 (Universal automatic computer)



programming - machine language

vacuum tube



ENIAC - first generation computer



1940-1956

vacuum tube



ENIAC - first generation computer

1940-1956

The Second Generation

- 1956-1963
- Transistor
  - Smaller
  - Less heat
  - Faster
  - More reliable
- Storage
  - Magnetic cores – primary memory
  - Magnetic disks – secondary memory
- Programming languages
  - Assembly language (mnemonics)
  - FORTRAN (1954)
  - COBOL(1959)
- Input: Punched card
- Output: Printout
- Example: IBM 1400, 7000 series, Control data 3600

Used primarily by business, university, government

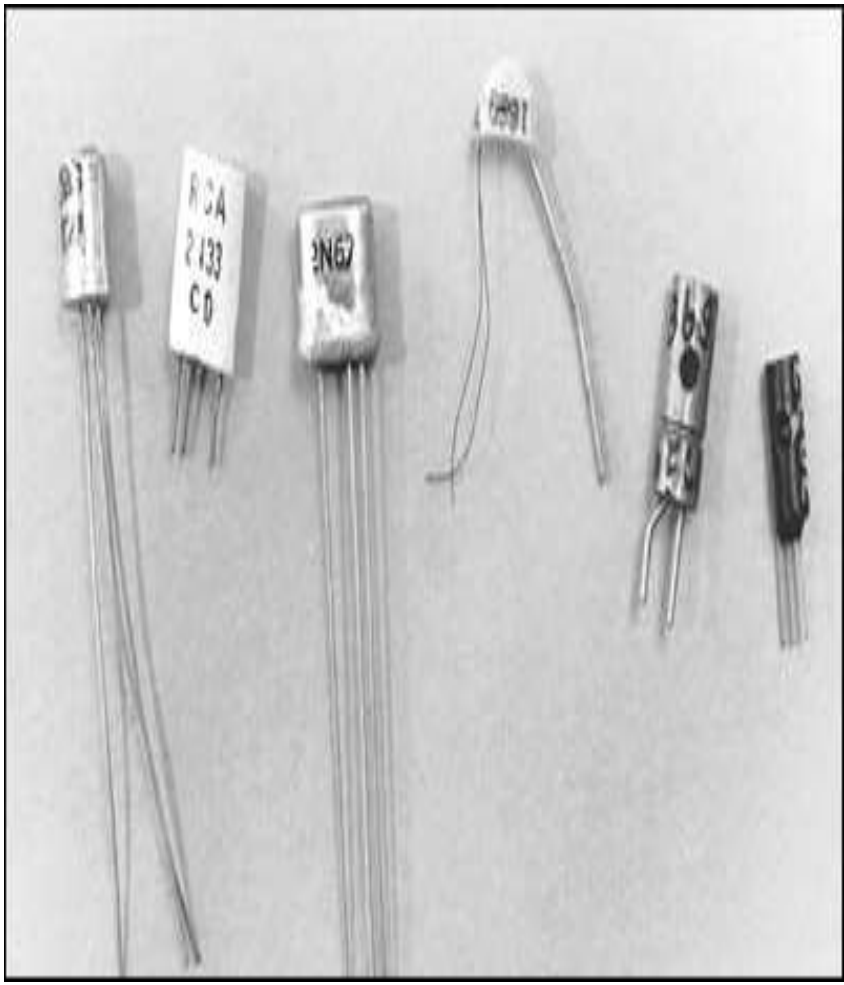


The Second Generation

- Computers became smaller
- Generate less heat
- Electricity consumption lower
- More reliable - less prone to hardware failure
- Better portability
- Assembly language used to program computers – programming became more time efficient

transistor





second generation computer

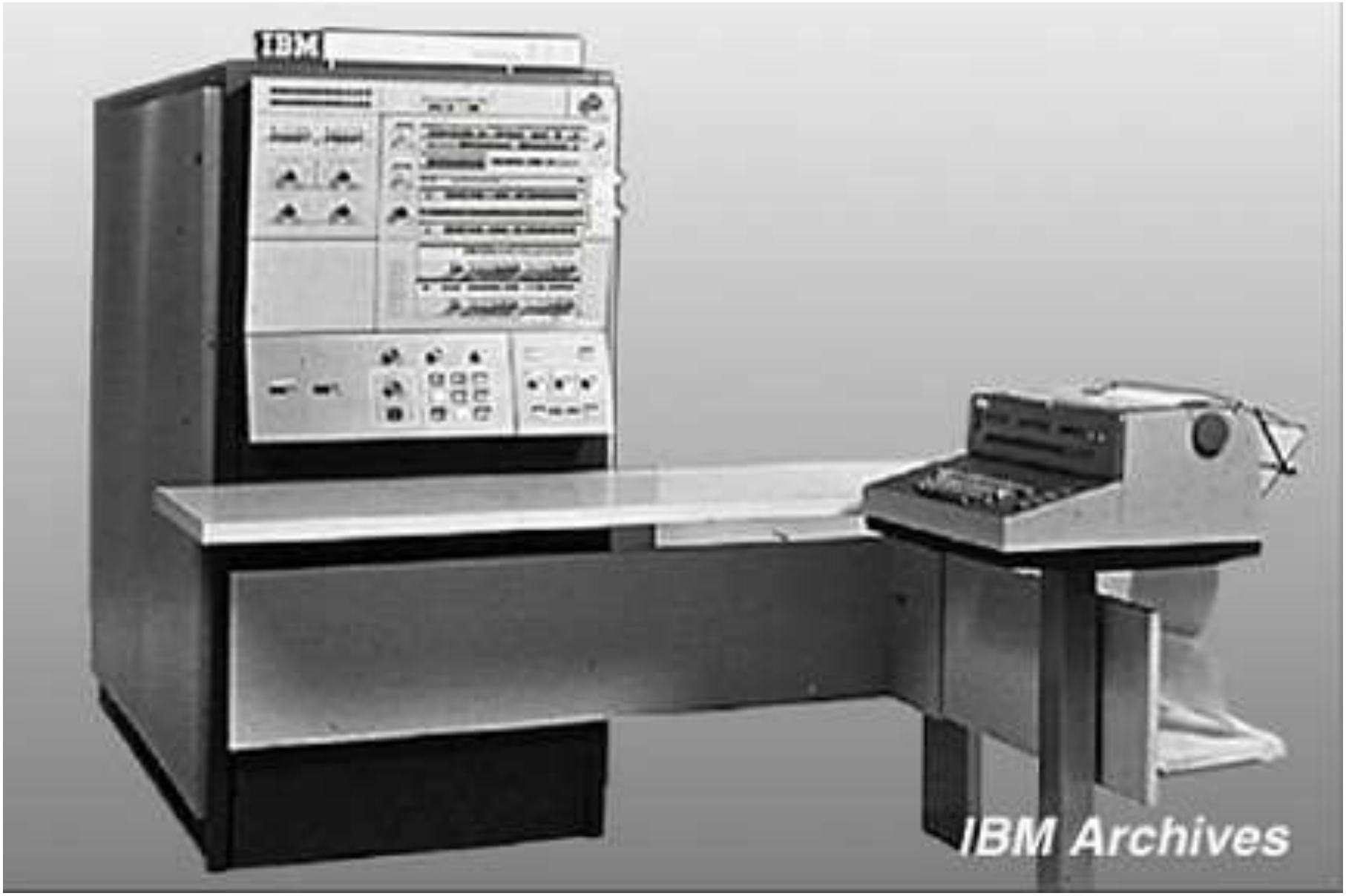
1956-1963

The Third Generation

- 1964- Early 1970
- Integrated Circuit
  - Electronic circuit on small silicon chip
  - Transistors and resistors fabricated on it

1. **Computers smaller, faster and more reliable**
2. **Power consumption lower**
3. **Storage capability upto Megabytes**
4. **High-level languages appeared**





third generation computer

1964-1970

The Third Generation

- Easily portable and more reliable
- Instead of punched cards and printout, for interaction with computer keyboard and monitors were used
- Size of computer was smaller compared to previous generation
- Example :IBM-360 series,ICL-1900 series,CDC-1700 Series

The Fourth Generation

- Large and Very Large Integrated Circuits(LSI & VLSI).i.e Microprocessor
- Portable Computer
- High Storage capability(upto Terabytes)
- Programming in High level Language
- UNIX OS,Dbase,C,ADA
- Example: Pentium, Power PC ,AMD, Apple Macintosh



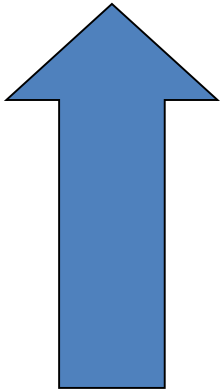
The Fourth Generation

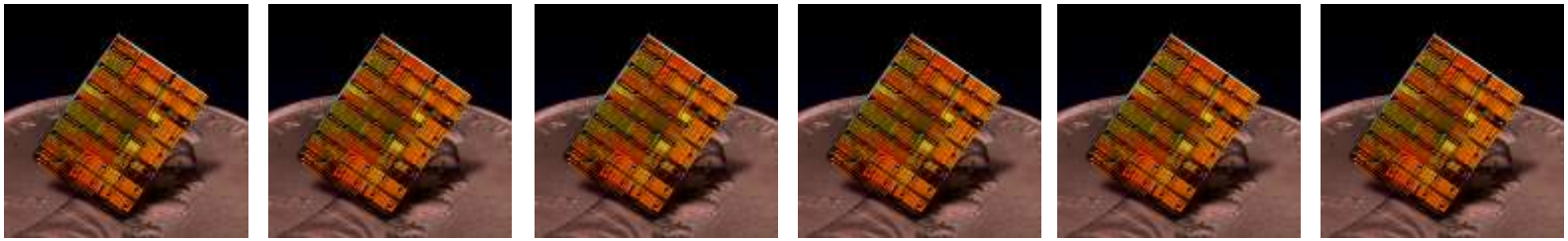
- Large Scale Integration (LSI): thousand of transistors are constructed on small chip
- Very Large Scale Integration (VLSI): hundreds of thousands of components on single chip
- Ultra – Large Scale Integration (ULSI): millions of components on single chip

1



1000





The Fourth Generation

- Powerful, compact, reliable and affordable – personal computer (PC)
- **Faster** random access main memory
- Hard disk became economical, smaller and bigger in capacity
- **Cheapest** among all the other generation
- Hardware failure is negligible, so **minimum maintenance** required
- Generate **negligible amount of heat**, hence do not required air conditioning
- **Interconnection with computer leads** to better resource sharing and communication, also developed pointing devices
- Disadvantage: Required **complex and sophisticated technology** for manufacturing of CPU and the other components.

The Fifth Generation

- Present and Beyond
- Based on artificial intelligence, uses the Ultra Large-Scale Integration (ULSI) technology
- Parallel processing method
- Portable and small in size.
- Input and Output: Trackpad (or touchpad), touchscreen, pen, speech input (recognize voice/speech), light scanner, printer, keyboard, monitor, mouse, etc.
- Examples:Desktops, laptops, tablets, smartphones, etc.











microprocessor

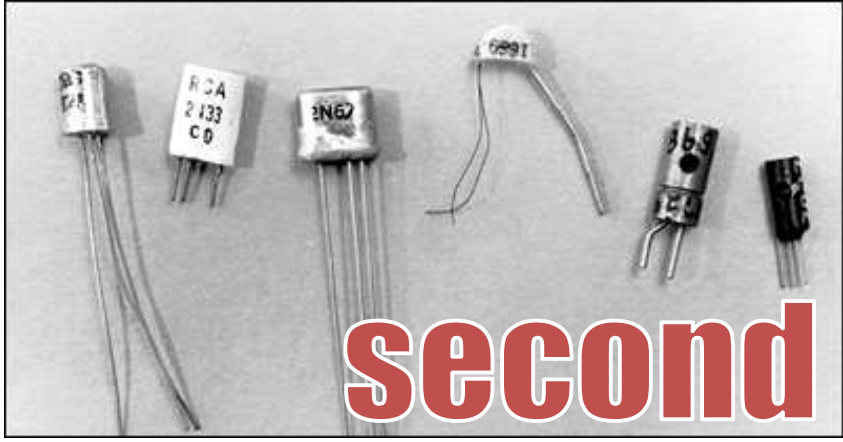
fifth generation computer

1989 - today

first

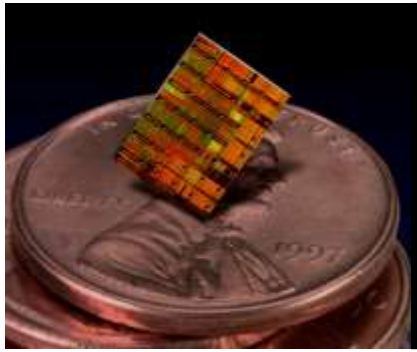
vacuum tube





second

transistor



third

integrated circuit





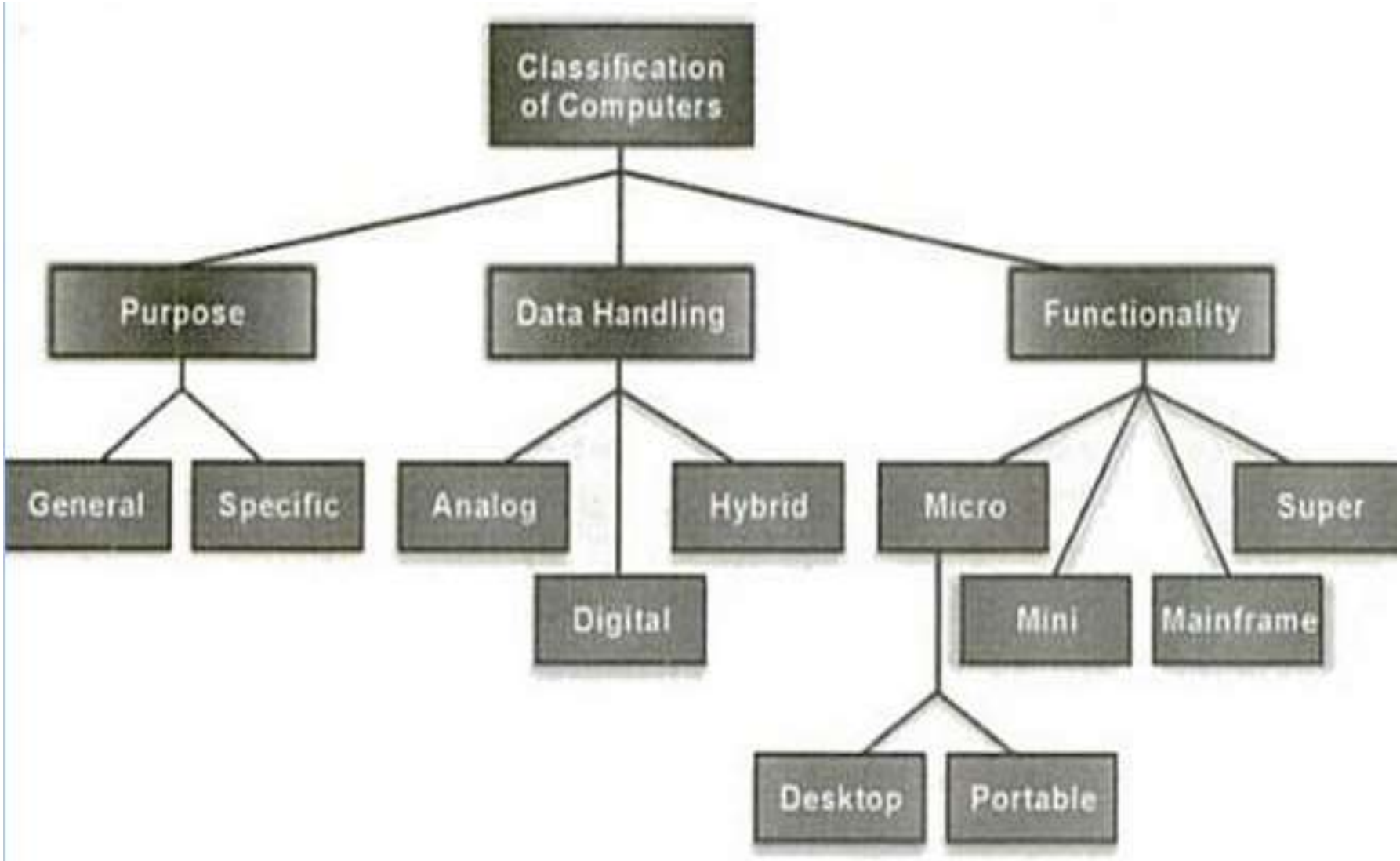
fourth

microprocessor

#	Computer Generations	Timeline	Hardware
1	First Generation of Computer	1940-1956	Vacuum Tubes
2	**Second Generation of Computer**	**1956-1963**	**Transistor**
3	Third Generation of Computer	1964-1971	Integrated Circuit (I.C.)
4	**Fourth Generation of Computer**	**1971-1980**	**Microprocessor**
5	Fifth Generation of Computer	1980- Till Now	Artificial Intelligence

	First Generation	Second Generation	Third Generation	Fourth Generation	Fifth Generation
Technology Used	Vacuum tube	Translator	IC	VLSI or microprocessor	Bio chip
Programming Language	Machine Language	Assembly Language	High Level Language	4GL (Problem Oriented)	Natural Language
Speed	Mili-Seconds	Micro second	Nano second	Pico second	More than Femto second
Primary Memory	Magnetic Core Memory	Magnetic Core Memory	Semiconductor Memory	Semiconductor Memory	-
Secondary Memory	Magnetic tape, Magnetic drum, etc.	Magnetic tape, Magnetic drum, etc.	Magnetic tape, Magnetic disk (Hard disk, Floppy disk)	Magnetic tape, Magnetic disk (CD, DVD, BD), Flash memory	-
Input Device	Punched Card	Punched Card	Keyboard	Mouse, Scanner	-
Output Device	Printing Device	Printing Device	Monitor	Touch screen, Printer	-
Cost, Size, Heat generation, Electrical Consumption	Very high	High	Low	Very Low	Not known
Speed, Capability	Very less	Less	Moderate	High	Much higher

Classification of Computers



ACCORDING TO FUNCTION OR PURPOSE

General Purpose Computers

- The general purpose computers are designed, developed, and manufactured for day-to-day tasks and operations
- Examples of General-purpose computers include Desktops[PC], Personal computers, Notebooks, and Tablets
- Inexpensive, compact , Cheap, Readily available in the market.
- Maintenance is less costly

Special Purpose Computers

- Special Purpose Computers are designed and developed for particular purposes and projects
- Examples of General purpose computers are Super Computers, Workstations, Servers, Personal Digital Assistants (PDAs),ATM
- Expensive, Bulky , Costly,Not available in local market.
- Maintenance costs are very much.

ACCORDING TO THE TYPE OF DATA HANDLED

1. ***ANALOG COMPUTERS.***
   were the first type to be produced. They measure continuous variable, physical quantities such as electrical potential, fluid pressure, mechanical motion etc.
   Do not deal with numbers. Measure continuous physical magnitudes.(temperature, pressure)
   Give approximate results, very fast and less accurate

Examples:
Speedometer of car,analog clock and thermometer, oscilloscope etc





2. *DIGITAL COMPUTERS.*

- handle information that can be counted. Uses digital circuits and are designed to operate on two states, namely bits 0 and 1.
- These bits can be combined to denote information such as numbers, letters, graphics, images and program instructions.
- Digital computers are suitable for complex computation and have higher processing speeds. They are programmable.
- Very accurate

Examples:

Desktop Computers, Lapto





3. *HYBRID COMPUTERS*

- is a combination of both analog and digital computer. Can handle both analog and digital data.
- Hybrid computers are well known for their ability to blend analog and digital features of computers. Advantages include the availability of both analog and digital computations within a single unit and an efficient processing speed





Application areas such as manufacturing, transportation, power systems and others

....ACCORDING TO SIZE & PROCESSING POWERS...

Micro Computers

- Micro Computer is a computer whose CPU (Central Processing Unit) is a microprocessor.
- All the components of a microprocessor are on a single integrated circuit chip.
- Micro computer can be categorized as the desktop, programmable and workstation.
- third generation computers.

- ***DESKTOP COMPUTERS***

  are the most common type of personal computer and are designed to fit conveniently on the surface of a desk or workspace.

  A desktop computer has a keyboard for input data, a LCD or CRT monitor to display information and Central processing unit tower contains storage, memory, different types of drives, such as, CD drive, hard drive, etc. A desktop computer is mainly used at home and office applications.







❑ *LAPTOP COMPUTER*

- also called a **notebook**, is a personal computer for mobile use.
- A laptop integrates most of the typical components of a desktop computer, including a display, a keyboard, a pointing device (touchpad or trackpad) and speakers into a single unit.
- Powered via an AC adapter, and can be used away from an outlet using a rechargeable battery.

Classification:

- ✓ **Desktop replacements**
- ✓ **Subnotebook**
- ✓ **Notebook**
- ✓ **Rugged**
- ✓ **Tablet PC**



FORMS:

- **Slate**
- **Convertible**
- **Hybrid**





❑ Handheld Computer

Are small, lightweight computers that are designed to be operated while held by hand or from a desk or tabletop.
Contain a processor, memory, I/O ports, and expansion slots.
Provide an operating system, user interface, and battery power.
Although extremely convenient to carry, handheld computers have not replaced notebook computers because of their small keyboards and screens.

Typical handhelds:

- Information appliance
- Smart phone/Mobile Phone
- Personal digital assistant (PDA)
- Personal Communicator









❑ EMBEDDED COMPUTERS

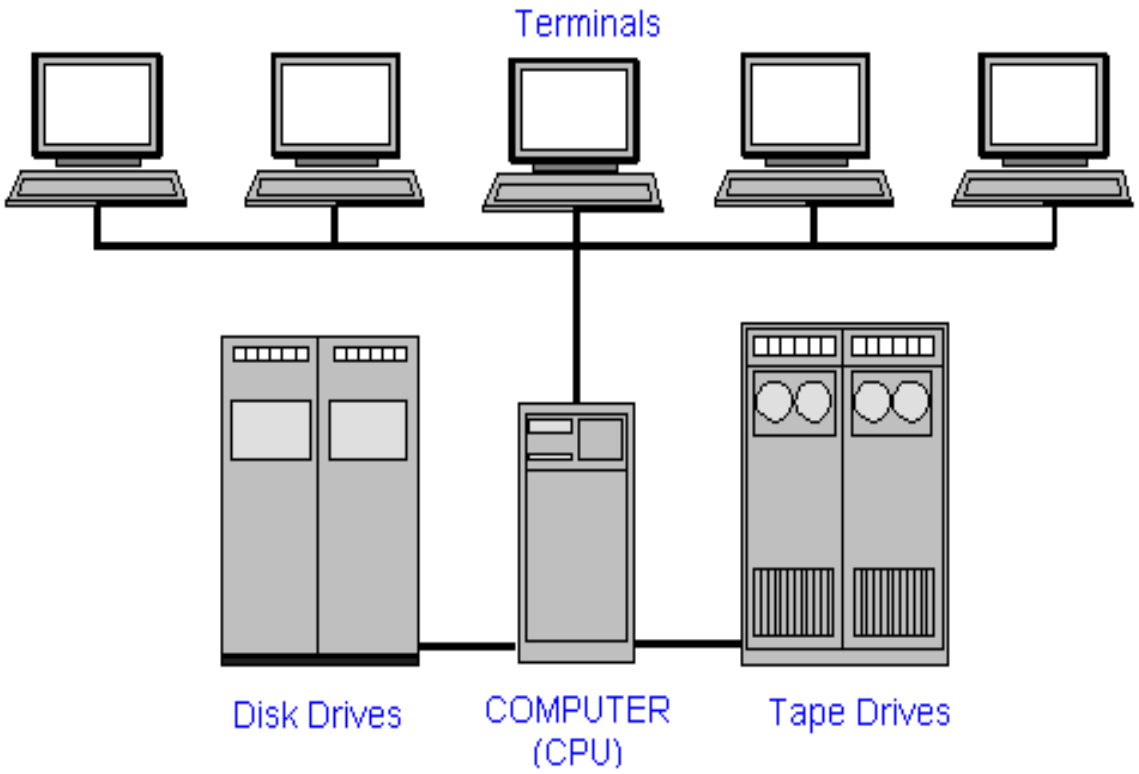
- ✓ “Computers on a chip”. A microprocessor or a microcontroller, which is programmed to perform a small number of tasks.
- ✓ It is embedded as part of a complete device often including electrical or electronic hardware and mechanical parts.
- ✓ are the “brains” behind many everyday mechanisms, such as wireless devices, cars, climate control systems, traffic signals, and washing machines, ticket machines at the subway, cameras, airplanes, cars, sewing machines, clocks, etc.





2. MINI COMPUTERS.

- Minicomputers are mid-range computers whose size, speed and capabilities lie somewhere, between those of a mainframe and a microcomputer.
- A minicomputer is a class of multi-user computers that lies in the middle range of the computing spectrum, in between the largest multi-user systems (mainframe computers) and the smallest single-user systems (microcomputers or personal computers).
- It is capable of supporting 4 to 200 users simultaneously. EX Network Server



3. MAINFRAME COMPUTERS

- they are very big in size and offer the maximum computing power. A large number of peripherals can be attached to them. They are generally used in large networks of computers with the mainframe being model point of the network. They used satellites for networking.
  - Powerful computers used mainly by large organizations for critical applications, typically bulk data processing such as online transaction processing system, extensive data storage and retrieval.
  - Most of the mainframe computers have the capacities to host multiple operating systems and can handle hundreds of users (can serve up to 50,000 users simultaneously)
  - Univac 1100/10, Univac 1100/60, Honeywell DSP 88/860, IBM 270/168 etc.

Mainframe Computer



4. SUPER COMPUTERS

- Fastest, costliest and most powerful. Generally operate at 4 to 10 times faster than the mainframe.They are most expensive of all the computers
- designed to work on a single problem at a time.
- Supercomputers are used for highly calculation-intensive tasks such as problems including quantum physics, weather forecasting, climate research, molecular modeling , physical simulations, aircraft and spacecraft designing, encrypting & decoding sensitive intelligence information
- Cray 1, Cray 2, Cray 3 perform 10 billions operation per second, Param, Cyber 810&830 etc.

Deep Blue computer play a chess game against Garry Kasparov in 1997, it is an IBM supercomputer that uses scalable parallel processing to solve complex problems. Deep Blue uses 256 processors working together to calculate between 50 and 100 billion chess moves in under three minutes.





Types of computer



Mini computer



Super computer



Micro computer



Mainframe computer

Size	Characteristics	Uses
super computers	Largest, fastest, powerful and most expensive. they also generate a lot of heat	Used for advanced scientific research such as nuclear physics.
Main frames	Less powerful and less expensive than super computers. They also have a large storage capacity	Used to handle all kinds of problems whether scientific or commercial. I.e. performing complex mathematical calculations. they are mostly found in banks, hospitals, airports etc.
Mini computers	Smaller and less powerful than the main frame	Used in scientific laboratories, research institutions, engineering plants and places where processing automation is required.
Micro computers	Smallest, cheapest and relatively least powerful. Uses a micro processor to process data. Examples: desktop, laptop and personal digital assistant (PDA)	Used to perform a variety of tasks including research, communication, banking, learning institutions, libraries etc.

Applications of COMPUTER

Uses of computer in e - commerce

- electronic mail
- video conferencing
- electronic shopping
- electronic banking



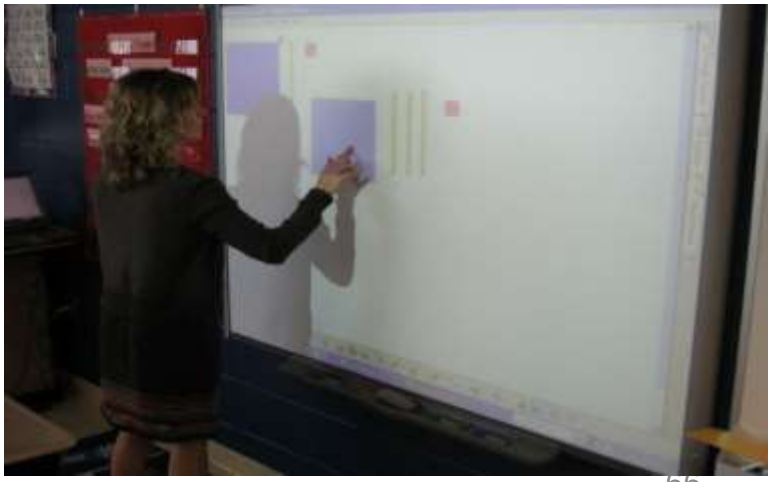


Education



- Better study moods
- most understandable and easier
- internet facility provides a lot of information from any corner of the world





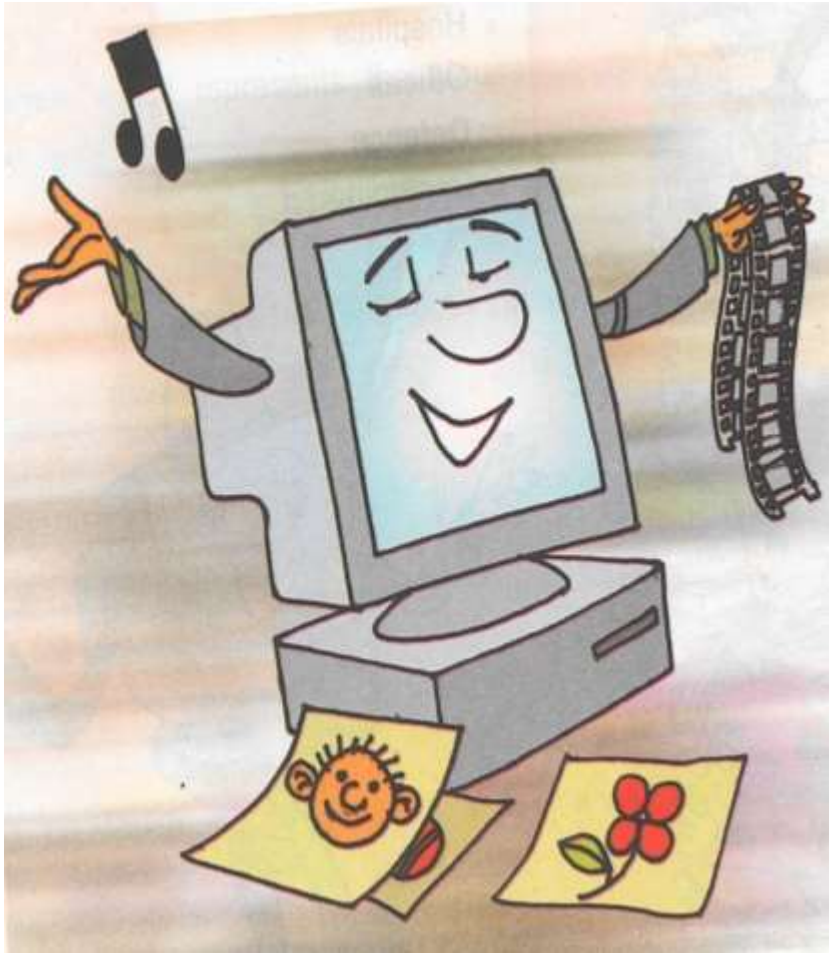
Medicine & Health care

- Patient monitoring
- patient records
- diagnosis
- hospital administration
- medical history records
- life support system





Entertainment



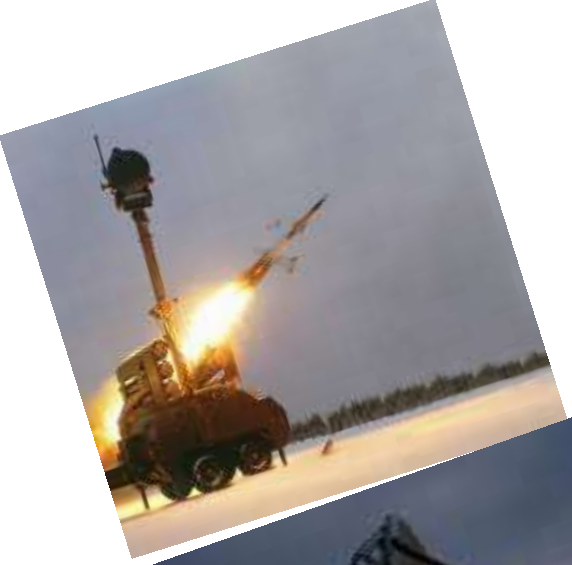
Engineering/Architecture/Manufacture







and many others …





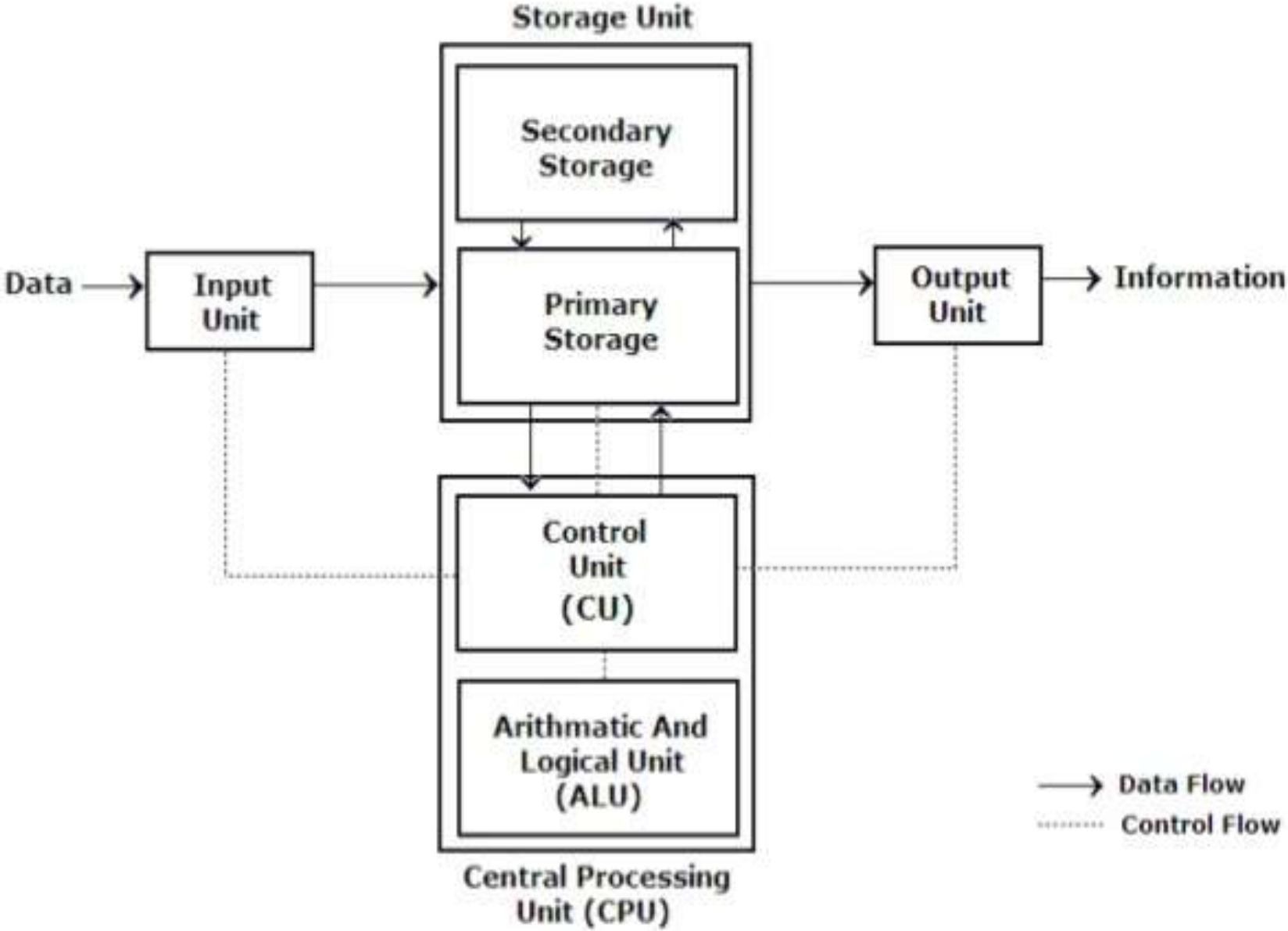








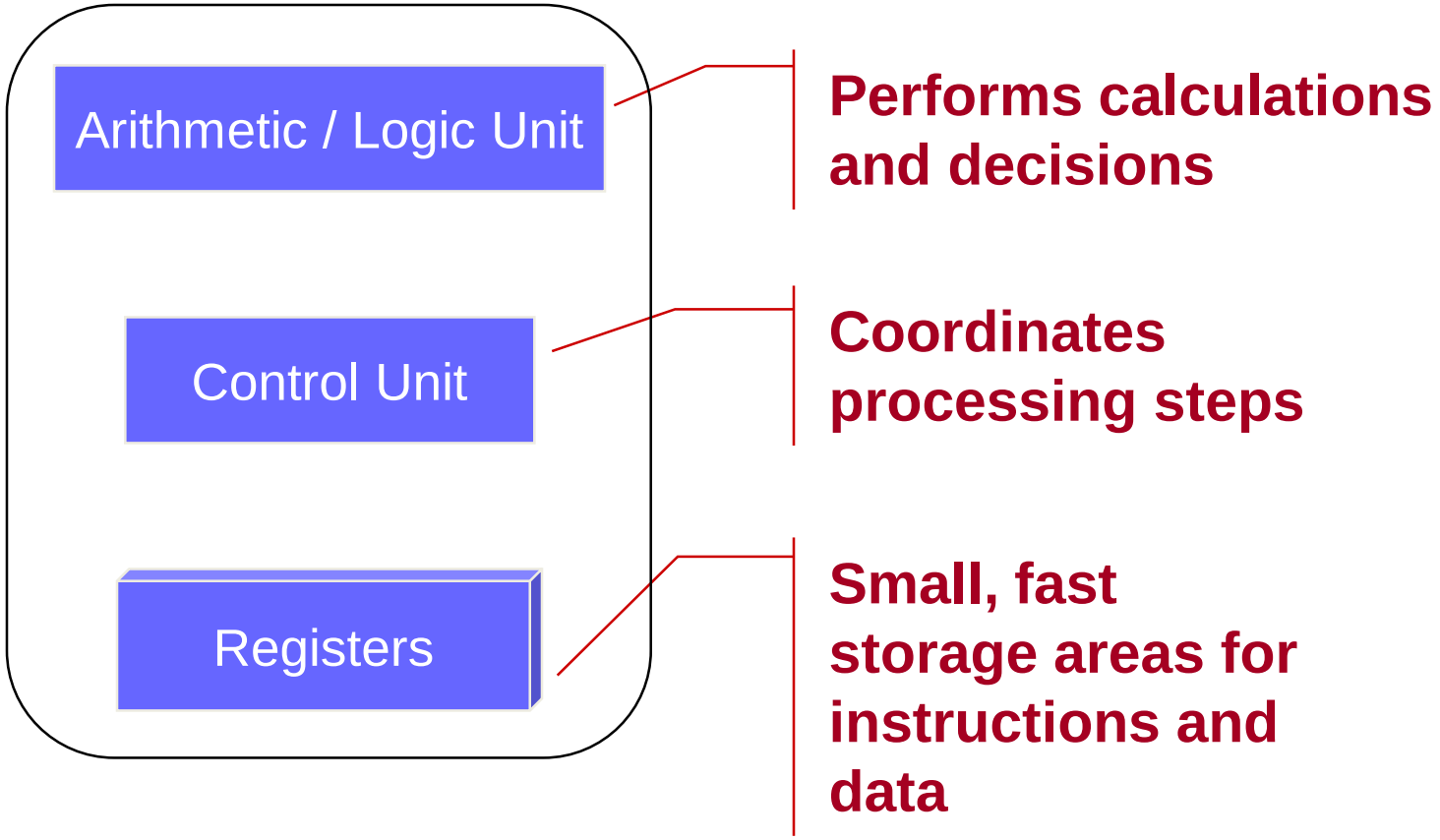
Block diagram of computer



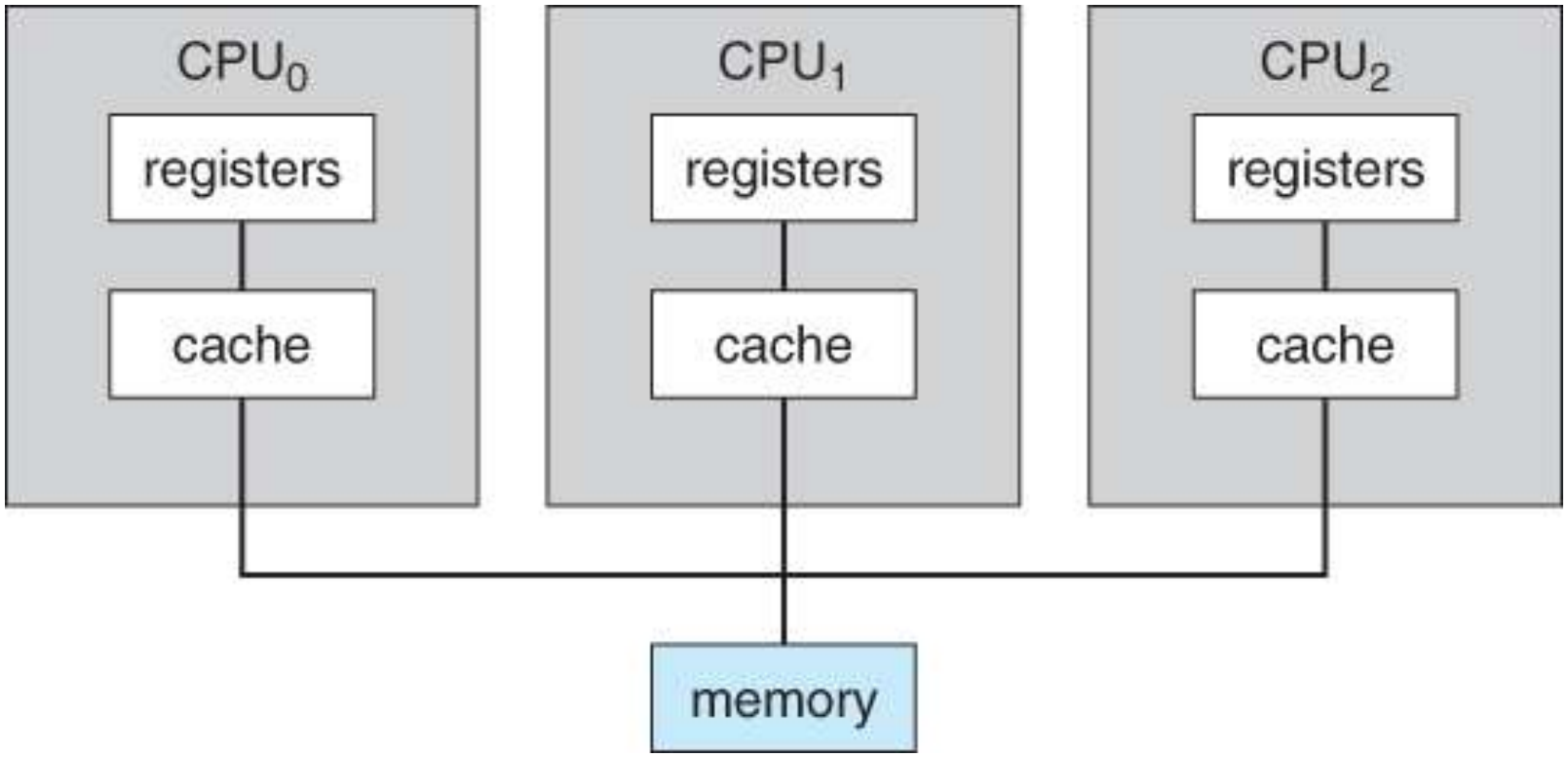
The CPU

- The **CPU** is a silicon chip that contains millions of tiny electrical components.
- The **CPU's** three main parts are:
  - Control Unit
  - Arithmetic Logic Unit (ALU)
  - Registers

Central Processing Unit (CPU)



Multiprocessor systems

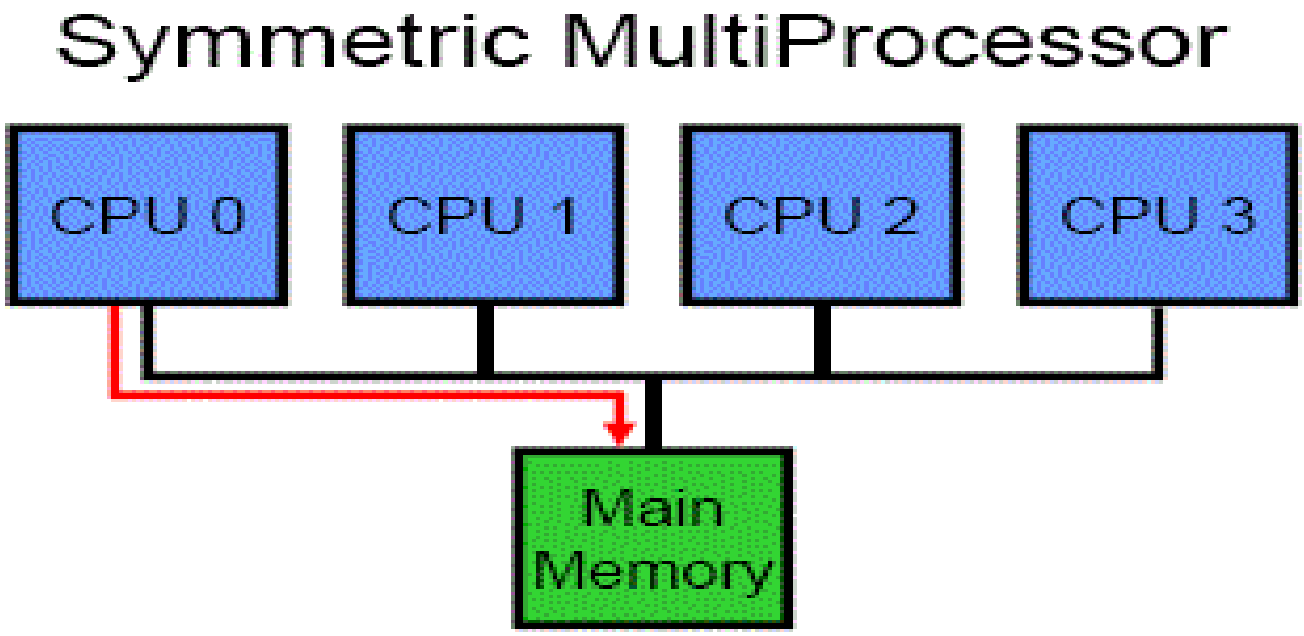


- A computer system which includes only one processor is called a single-processor system.

- Computer system that include more than one processor are called multiprocessor systems.

Multiprocessor systems

- Two popular architectural approaches for building multiprocessor computer systems are:
  - Symmetric Shared Memory Multiprocessor
  - Asymmetric Shared Memory Multiprocessor



- Common pool of memory connected with high speed bus
- All processor can access all these memory

Asymmetric Multiprocessor

- One processor act as a controller termed as *master processor*, which controls memory unit and other processors.
- Rest of the processor are termed as *slave processor*
  - The master has access to whole memory *but*

Slave has access to limited memory